AMRITA VISHWA VIDYAPEETHAM
DEEMED TO BE UNIVERSITY UNDER SECTION 3 OF UGC ACT, 1956
COIMBATORE CAMPUS

INSTITUTION'S INNOVATION COUNCIL
(Ministry of Education Initiative)

GENESIS

Genesis Ideathon: Final Presentation

Team ID: *GEN26-A0H-I180*

Team Name: *DRAGON BALL Z*

Domain: *DOMAIN 8: Space, Aerospace, Mobility & Next-Gen Infrastructure*

Amrita Vishwa Vidyapeetham, Coimbatore Campus

Date: 11th October 2026

Presentation Outline

1. Problem Statement
2. Problem Understanding
3. Existing Gaps & Challenges
4. Our Proposed Solution
5. Technical Approach
6. Innovation & Uniqueness
7. Feasibility & Scalability
8. Cost for Product
9. Use Cases
10. Impacts and Benefits
11. Sustainable Development Goals
12. Individual Member Contribution

Problem Statement

eVTOL Instability in transition process due to various factors.

Problem Understanding

- eVTOL commercial operators and future passengers face this critical risk, as the aircraft is highly vulnerable to losing control or drop altitude during the shift between flight modes.
- This instability arises from a critical clash of aerodynamic and mechanical forces,As the aircraft shifts modes, it enters an aerodynamic "Stall Gap" ,which means traveling too fast for steady rotor hover but too slowly for the wings to generate stable lift, resulting in turbulent buffeting.
- This leads to catastrophic safety risks, high maintenance costs and reduced aircraft range. For passengers, it directly affects the reality of safe, reliability, and smooth urban air mobility.

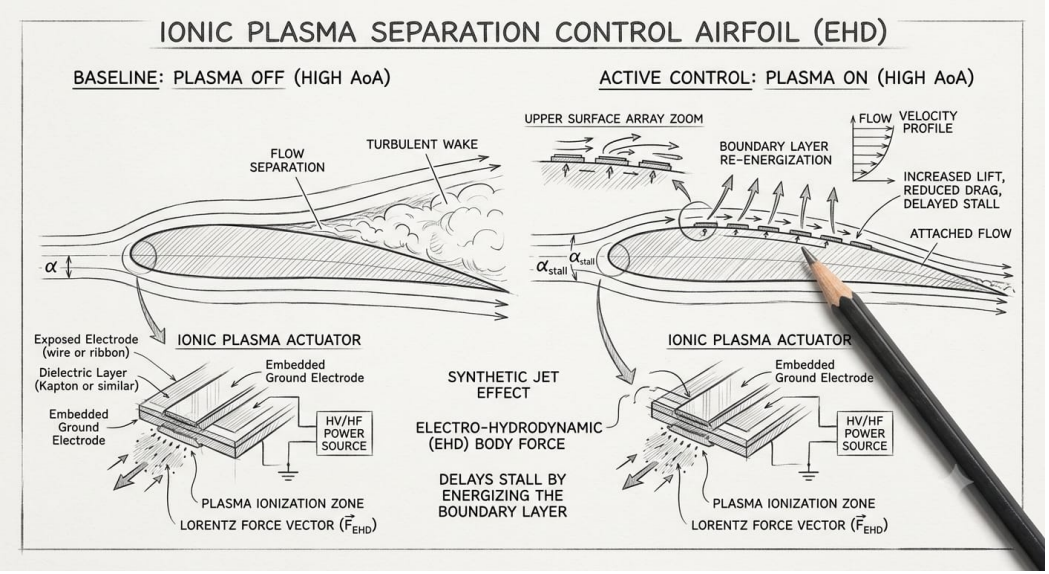
Existing Gaps & Challenges

- Traditional eVTOLs rely on slow, mechanically tilted rotors to transition to forward flight, forcing the aircraft into a dangerous aerodynamic "dead zone" where there is no air flow attached to the airfoil.
- Current structures lack a lightweight and instantaneous secondary propulsion mechanism capable of accelerating the aircraft into a gentle glide without adding heavy, dead-weight backup battery packs or more motors.Normal lithium-ion batteries cannot withstand the high dual-load power spike demanded by simultaneous hover and forward propulsion without choking the aircraft's power grid and reducing motor control responsiveness.
- For stability Adding more motors reduces flight time because the extra weight of the hardware and the added wind resistance create a penalty that drains the battery faster.

Our Proposed Solution

- An Integrated electric Ionic Plasma Wing Booster replaces heavy, complex tilting mechanical motors with air-breathing microwave plasma channels embedded directly on the wing structure to act as a forward thrust booster during the critical transition phase
- A high-power Supercapacitor Bank dumps an isolated megawatt-scale electrical voltage to ignite the wing's plasma segments, instantly providing the acceleration need to the aircraft for a stable wing-borne glide and this will maintain a small amount of air flow above the wing which prevents stall.
- Passenger Safety and longer Operational Range,silent, lower-maintenance, highly reliable aircraft by eliminating heavy mechanical joints.

Technical Approach



A silent, steady airflow is created when a high voltage applied to a sharp electrode ionizes surrounding air molecules, accelerating them toward an opposite electrode and dragging neutral air along through momentum transfer.this generates the lift required

Tech Stack

Frontend: react.js

Backend: phyton

Database: influxDB

AI/ML, APIs, or Hardware: power electronics

Innovation & Uniqueness

- A electric ion plasma Propulsion System, Traditional eVTOL solutions rely strictly on rotating mechanical parts or moving aerodynamic surfaces to handle transition. Our design introduces, a zero-moving-part ionic plasma booster embedded into the wing's airfoil skin, completely eliminating mechanical failure during mode-switching and in high distrubance situations.

- This approach introduce a new concept into aviation to prevent stall, which is to maintain a small amount of air flow above the airfoil to keep generating lift.we are doing this without adding much weight like another motor which in turn reduces it time of flight and its size.we are making this compact and efficient.

- The key difference is that we are integrating this plasma ionic propulsion onto the wings,which is lighter than a motor. So,the wing structure doesn't have to be made to withstand high loads.this reduces the aircraft weight and it's size making it a more usable and much safer.

Feasibility & Scalability

- **Feasibility:** This concept is highly achievable today because it combines existing, commercially proven hardware with validated aerospace plasma technology. We also take Silicon-Carbon batteries, which are in development and standard high-voltage modules, requiring no raw materials discovery.the entire power propulsion architecture can be fully prototyped and benchmarked on a laboratory workbench.

- **Scalability:** The ion plasma tiles can scale upto lift heavy cargo or multi-passenger aircraft.This system can expand to completely replace heavy mechanical wing flaps and spoilers with plasma-steered airflow.The ionization system can adaptively scale power to maintain thrust in thin air or high altitudes.

- **Risks:** Plasmic distrubance and battery thermal stress during transitionand. The high physical swelling of silicon anodes during fast discharge,if si is used.

Cost for Product (Optional)

- Development or hardware cost: 1.8+ crore Rupees
- Hosting or maintenance cost: NA
- Estimated total: 2.5 crore Rupees

Use Cases (compulsorily 3 use cases must be there)

- Use case 1: Immediate emergency response in hard to reach terrains.
- Use case 2: noise less and thermally undetectable military applications.
- Use case 3: no-slip condition in advanced hypersonic missile and high speed aircraft surface.

Impacts and Benefits

- Maximized Transition Safety Smooth Flights: Passengers experience zero terrifying altitude drops or heavy mechanical vibration during the transition phase, while commercial operators benefit from low-maintenance solid-state wings and a highly responsive, unchokable flight power grid.
- low Emission Urban Aviation with Minimal Manufacturing Footprint: We eliminate global warming emissions via air-breathing ionic propulsion while leveraging eco-friendly, carbon-scaffold silicon batteries. This removes the heavy recycling toll of oversized, dead-weight backup batteries and drastically lowers vehicle maintenance costs.
- forty five percentage range extension and zero voltage sag: eliminating heavy mechanical tilting assemblies and standard graphite anodes. alongside a hundred percent reduction in transient voltage drop during high-power transition bursts.

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
SDG 1	Target [9.4]	Industry, Innovation, and Infrastructure	Upgrades aviation infrastructure by replacing heavy mechanical parts with a resource-efficient, zero-emission ionic plasma drive.
SDG 2	Target [11.2]	Sustainable Cities and Communities	Provides safe and whisper-quiet urban air mobility, allowing cities to adopt clean vertical transit that bypasses traditional road congestion.

Individual Member Contribution

Member	Contribution or responsibility
Balaji V	Researched about current eVTOL designs, limitations and the identified gap.
S Nithin	Researched about ionic plasma propulsion
Pranav R	Researched about Si-C battery
Madhavan N	Researched about advantages and disadvantages of our idea

Yukti ID

IR2026-1051062